Open Science, Open Source and Code Sharing

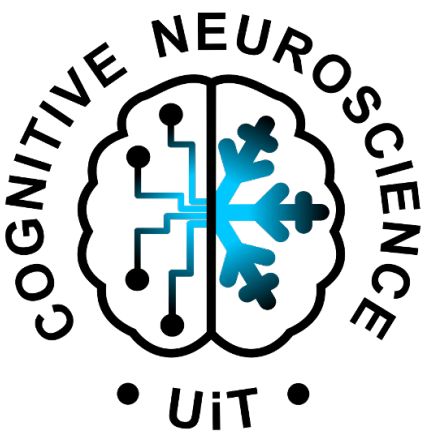
PSY-3035 · Autumn 2026

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2026-10-26



Open Source

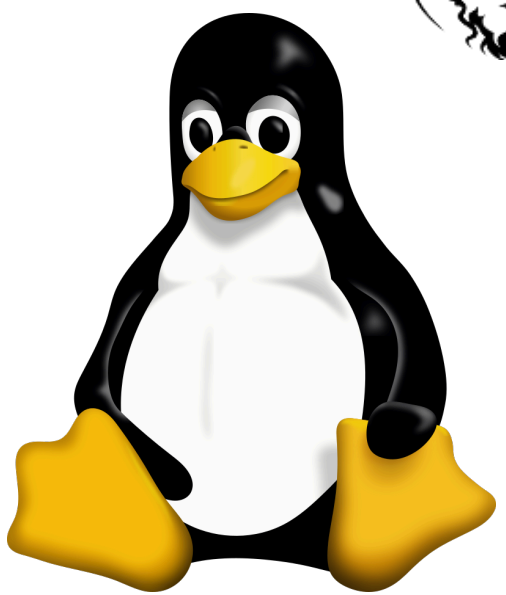


open source initiative®

FREE SOFTWARE FOUNDATION

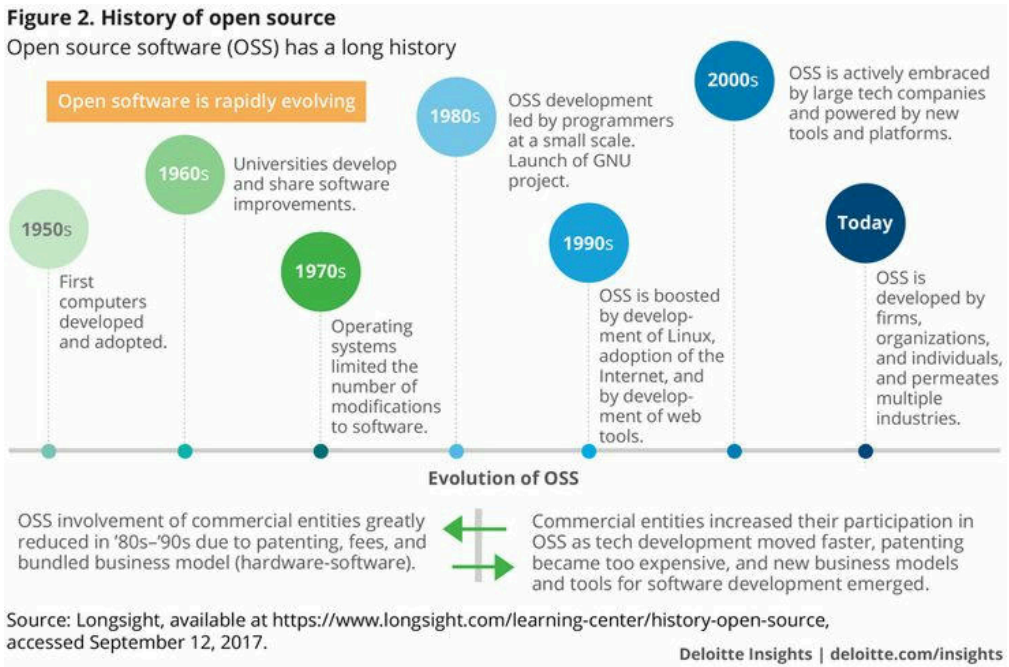






Logos: Open Source Initiative; Free Software Foundation; GNU head by Aurelio A. Heckert (CC BY-SA 2.0); Tux by Larry Ewing (made with GIMP); Python Software Foundation

A short history of Open Source



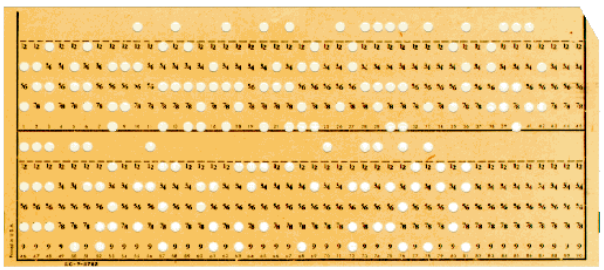


Linus Torvalds and Richard Stallman

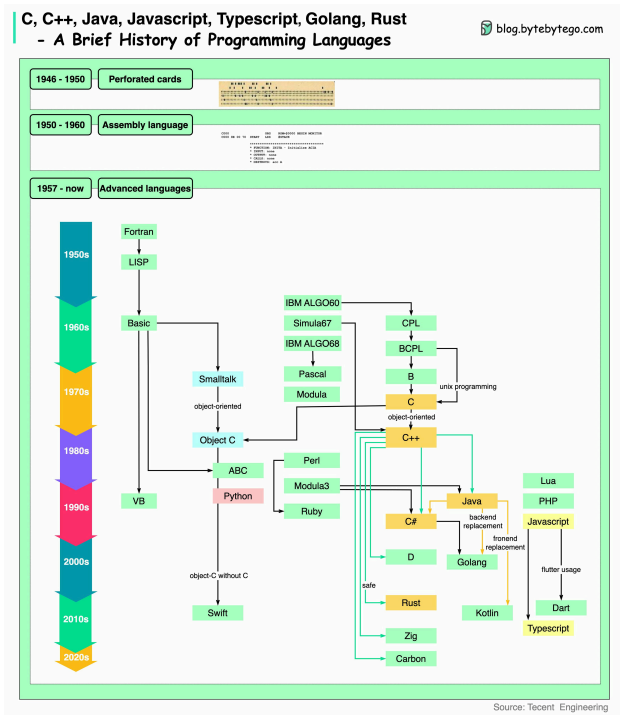
1950s–60s	**Sharing is normal**: code comes with the hardware, users and universities swap it
1969	**Software becomes a product**: IBM sells it separately, the source gets closed
1983–89	**Free software**: Stallman starts GNU, the FSF and the GPL ("free as in freedom")
1991	**Linux**: a student's "hobby" OS, built by volunteers over the internet
1998	**"Open source"**: OSI founded
2000s–today	**Mainstream**: Linux runs servers, Android, supercomputers, …

Timeline: Deloitte Insights, based on Longsight (2017); photos via Tech Source; Williams (2010) *Free as in Freedom 2.0*, FSF (static.fsf.org/nosvn/faif-2.0.pdf)

History of programming languages

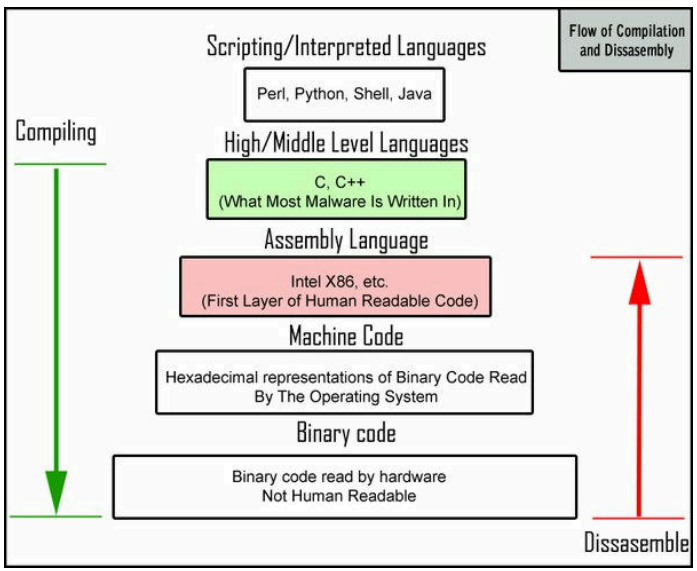


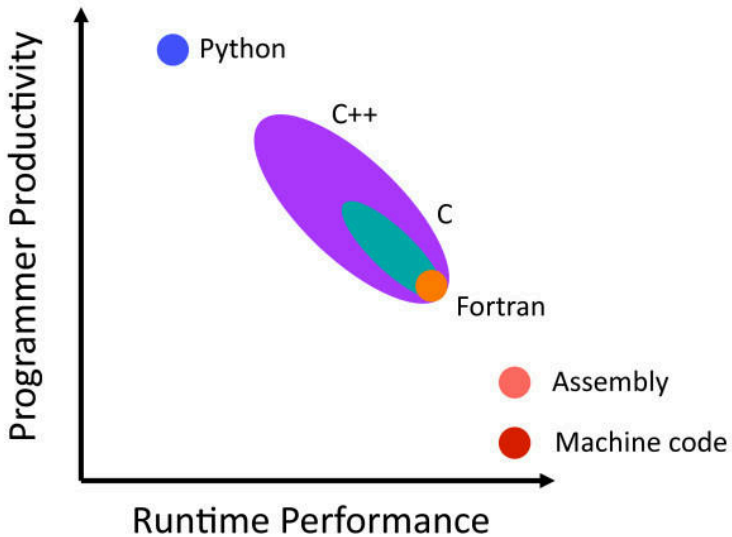




Punched card: Punch Card Gallery; stack of punched cards: Computer History Museum; language history: A. Xu, ByteByteGo (blog.bytebytego.com)

The *source* in Open Source

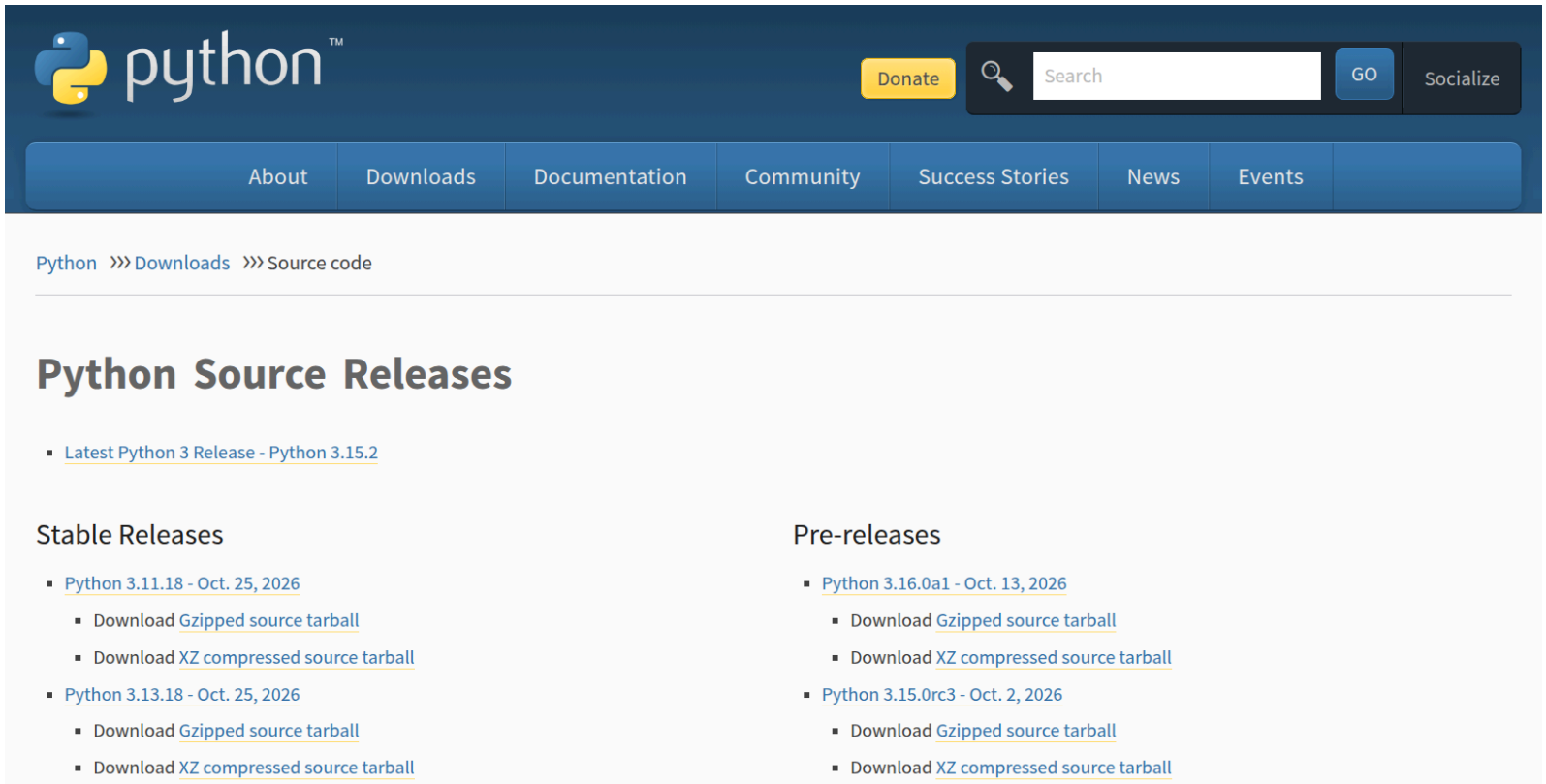




⟶ **source code** is the human-readable version of a program; open source means *you get that version*

Left: via Quora; right: via ResearchGate

Python is Open Source



Screenshot: python.org/downloads/source (26.10.2026); source code also on github.com/python/cpython

Open Source needs a licence

code that is public is not automatically "open"

- **No licence = all rights reserved:** others may *read* your code, but not legally reuse, modify or redistribute it
- **Permissive** (MIT, BSD, Apache 2.0): do anything, just keep the copyright notice
- **Copyleft** (GPL): derived work must stay open under the same licence (Stallman's idea)

- **Data, text, figures:** Creative Commons
  - → **CC BY**: reuse with attribution
  - → **CC0**: public domain
- Add a `LICENSE` file to every repository

Open Source Initiative (opensource.org/licenses); creativecommons.org/licenses

Choosing a licence



Choose an open source license

An open source license protects contributors and users. Businesses and savvy developers won't touch a project without this protection.

Which of the following best describes your situation?

I need to work in a community.

Use the **license preferred by the community** you're contributing to or depending on. Your project will fit right in.

If you have a dependency that doesn't have a license, ask its maintainers to **add a license**.

I want it simple and permissive.

The **MIT License** is short and to the point. It lets people do almost anything they want with your project, like making and distributing closed source versions.

Babel, **.NET**, and **Rails** use the MIT License.

I care about sharing improvements.

The **GNU GPLv3** also lets people do almost anything they want with your project, *except* distributing closed source versions.

Ansible, **Niri**, and **uBlock Origin** use the GNU GPLv3.

What if none of these work for me?

My project isn't software.

There are licenses for that.

I want more choices.

More licenses are available.

I don't want to choose a license.

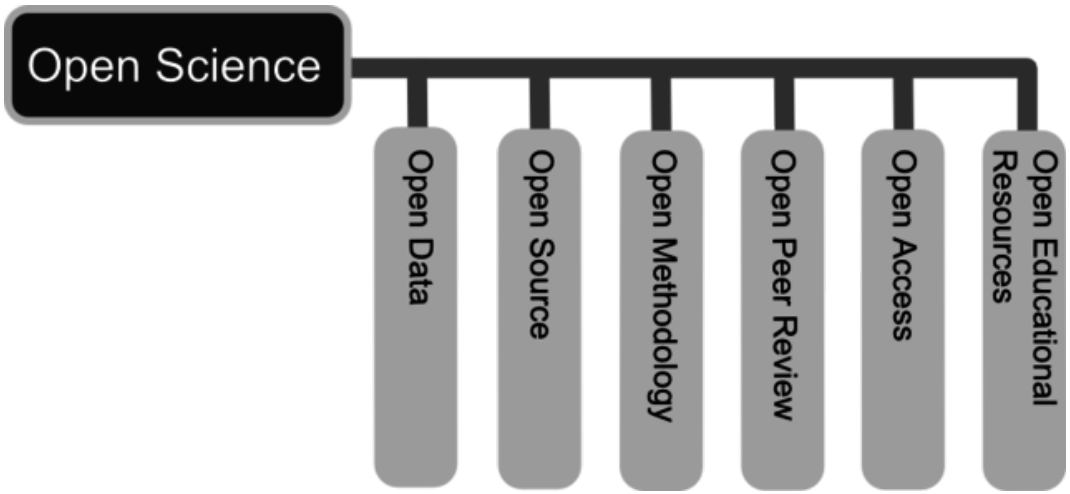
Here's what happens if you don't.

- Help choosing: choosealicense.com

Screenshot: choosealicense.com (26.10.2026)

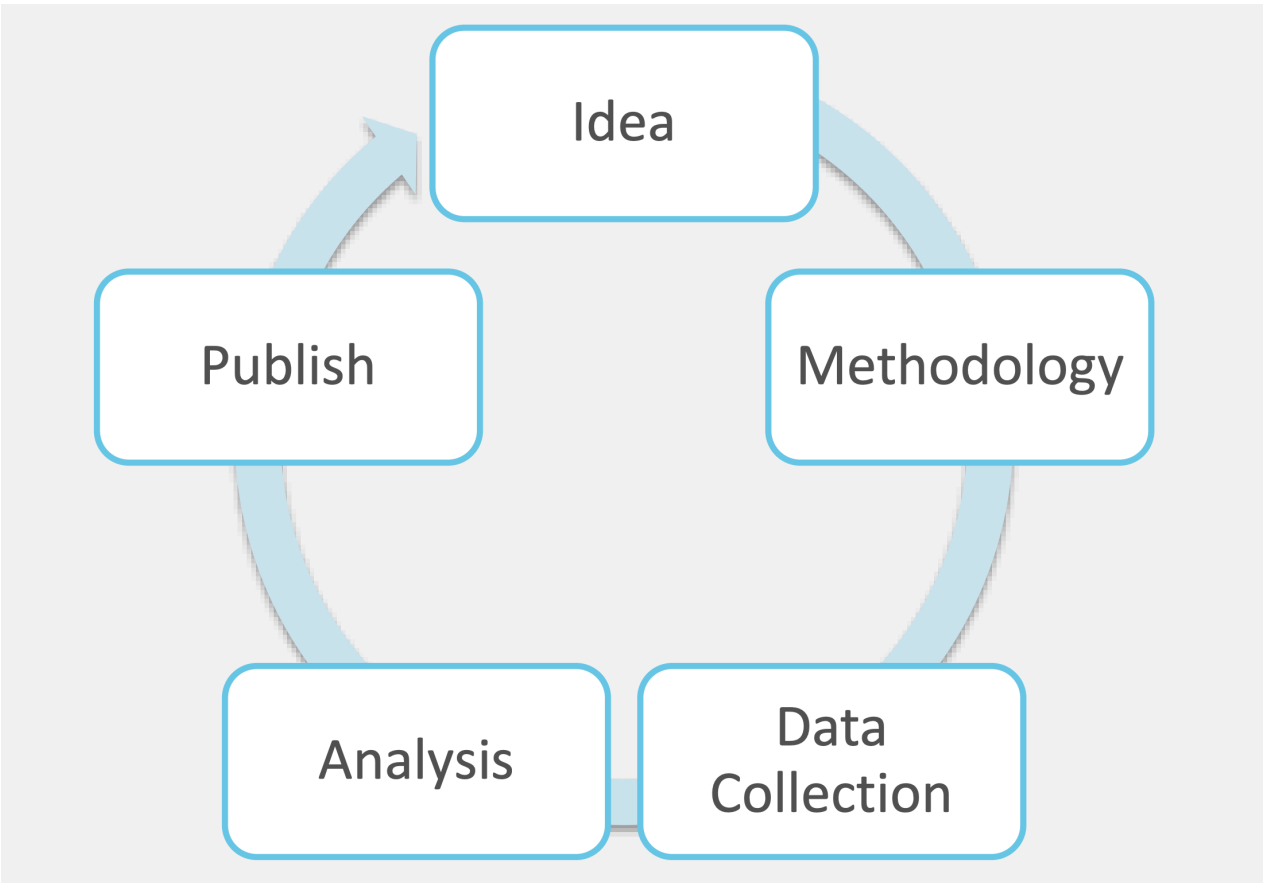
Open Source and Open Science?

> The movement to make scientific research, data and dissemination accessible to all levels of an inquiring society.

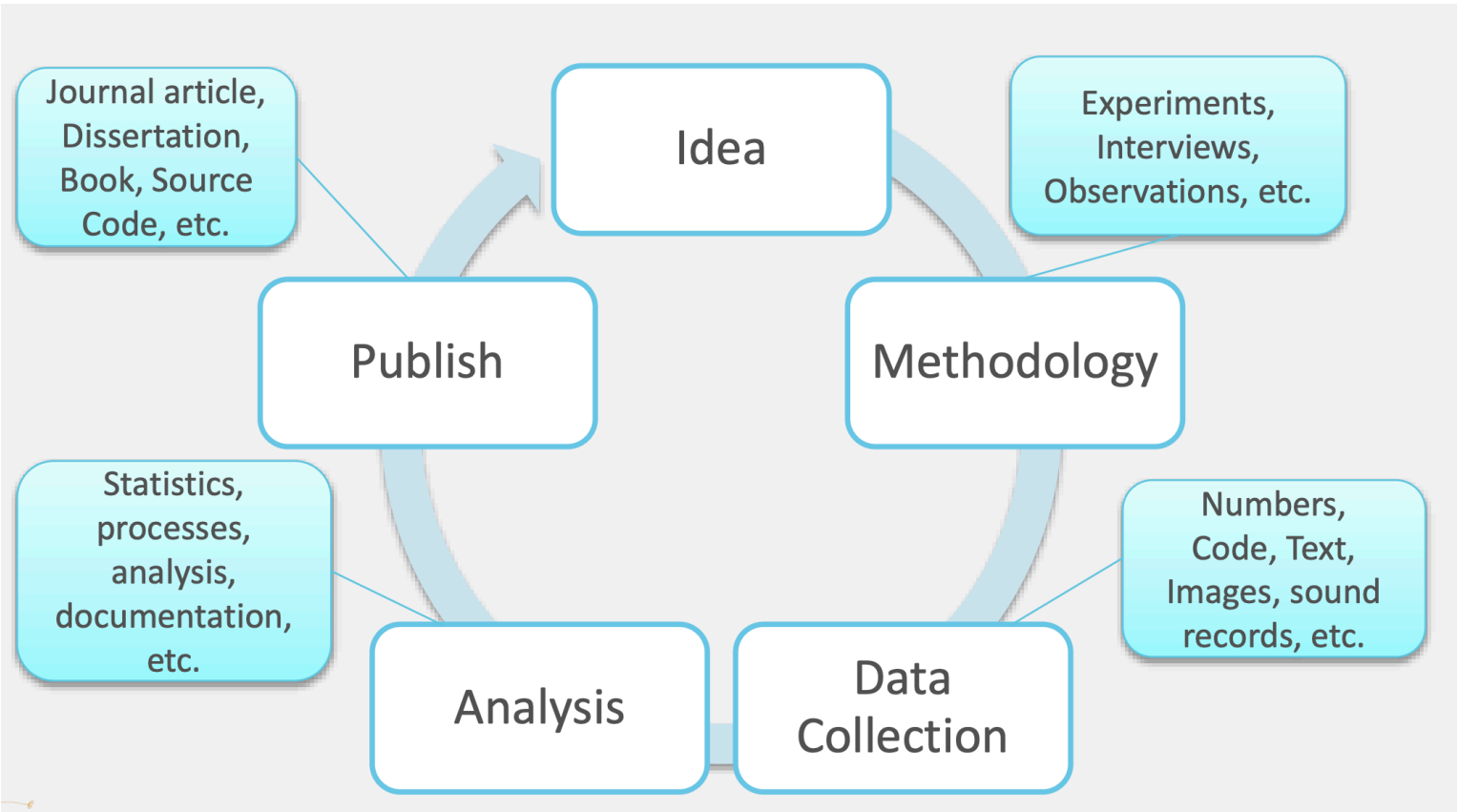


FOSTER Open Science definition (fosteropenscience.eu); Figure: Andreas E. Neuhold, “Open Science – Prinzipien”, CC BY 3.0, Wikimedia Commons

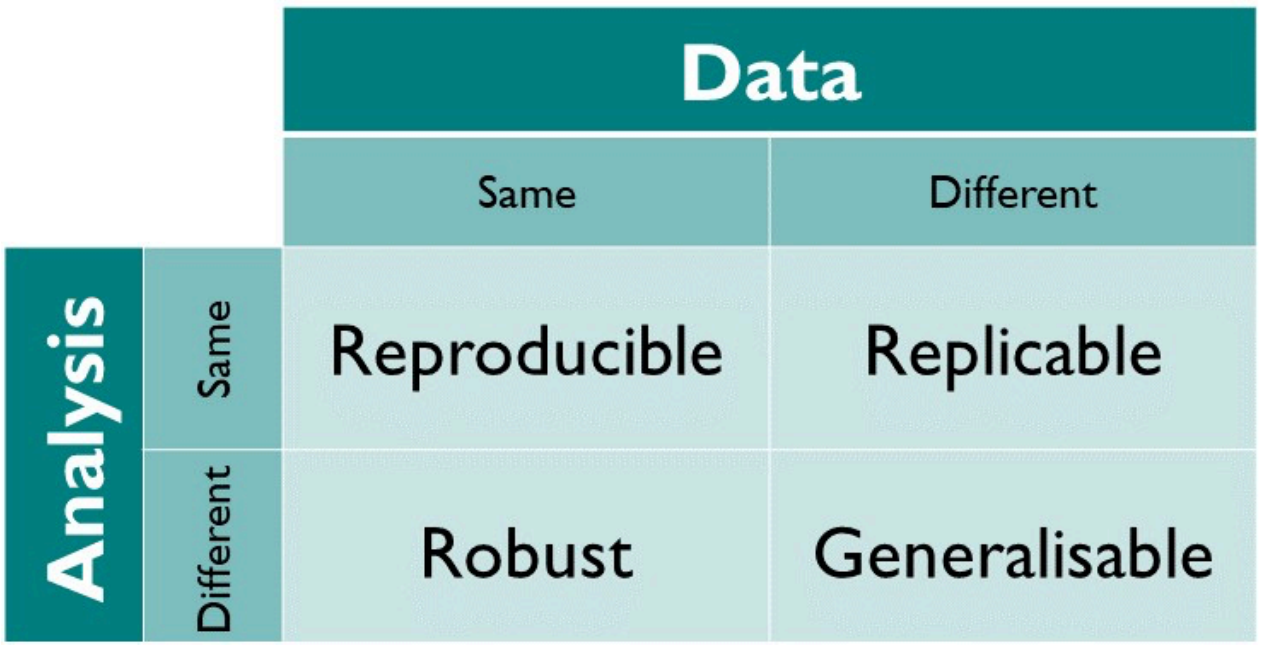
A rough sketch of the scientific process



A rough sketch of the (open) scientific process

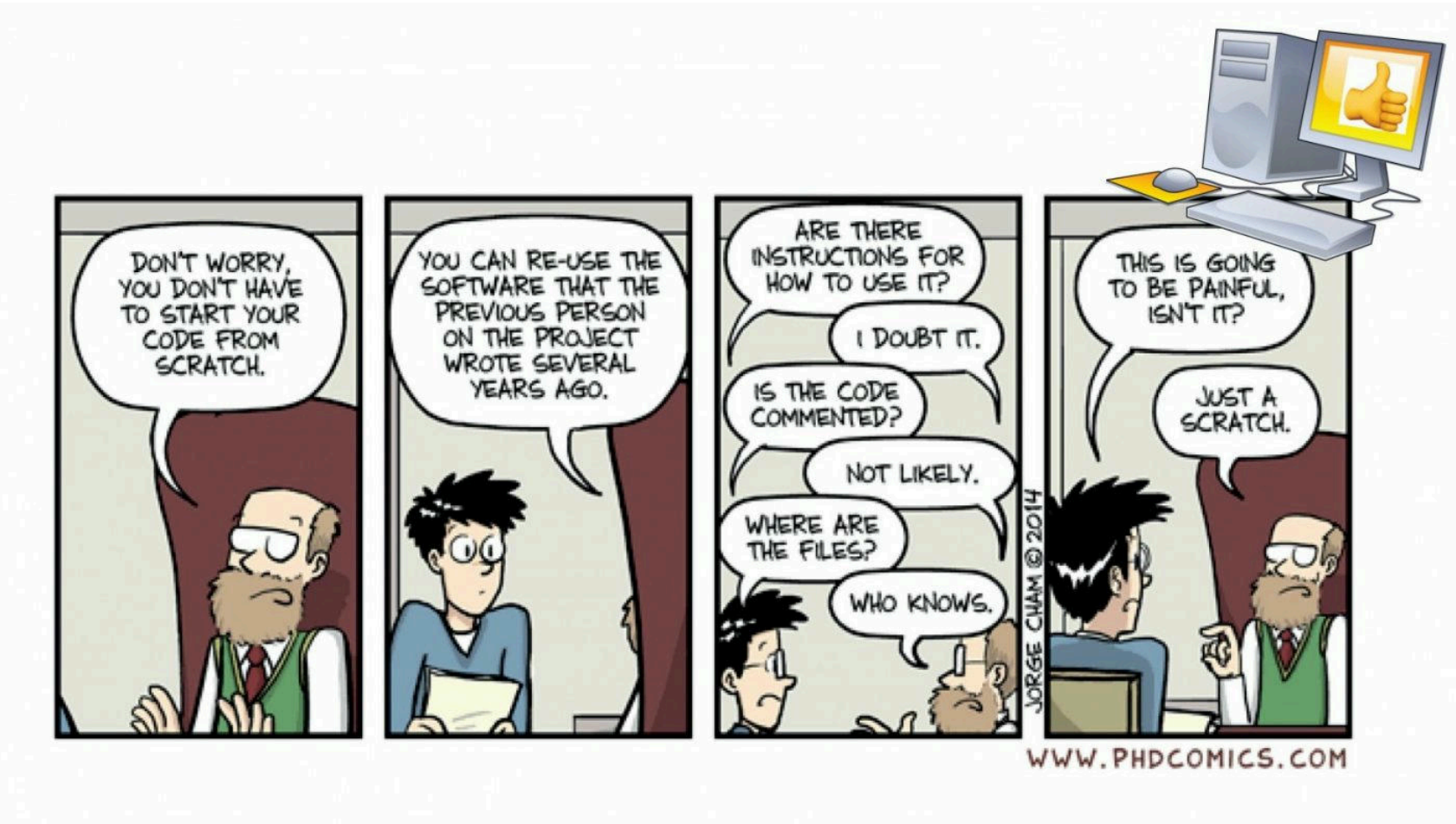


Reproducibility vs. Replicability



The Turing Way Community (2022). The Turing Way, CC BY 4.0. doi:10.5281/zenodo.3233853

Reproducibility: Sharing code with your future self



Is psychology computationally reproducible?

same data ⟶ same result?

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articles in *Cognition* with open, reusable data: at least one value could **not** be reproduced, even with help from the authors (another 11 needed that help)

Hardwicke et al. (2018)

20 / 62

Registered Reports in psychology: main results reproduced from the shared data and code (only 35 shared both)

Obels et al. (2020)

29 teams

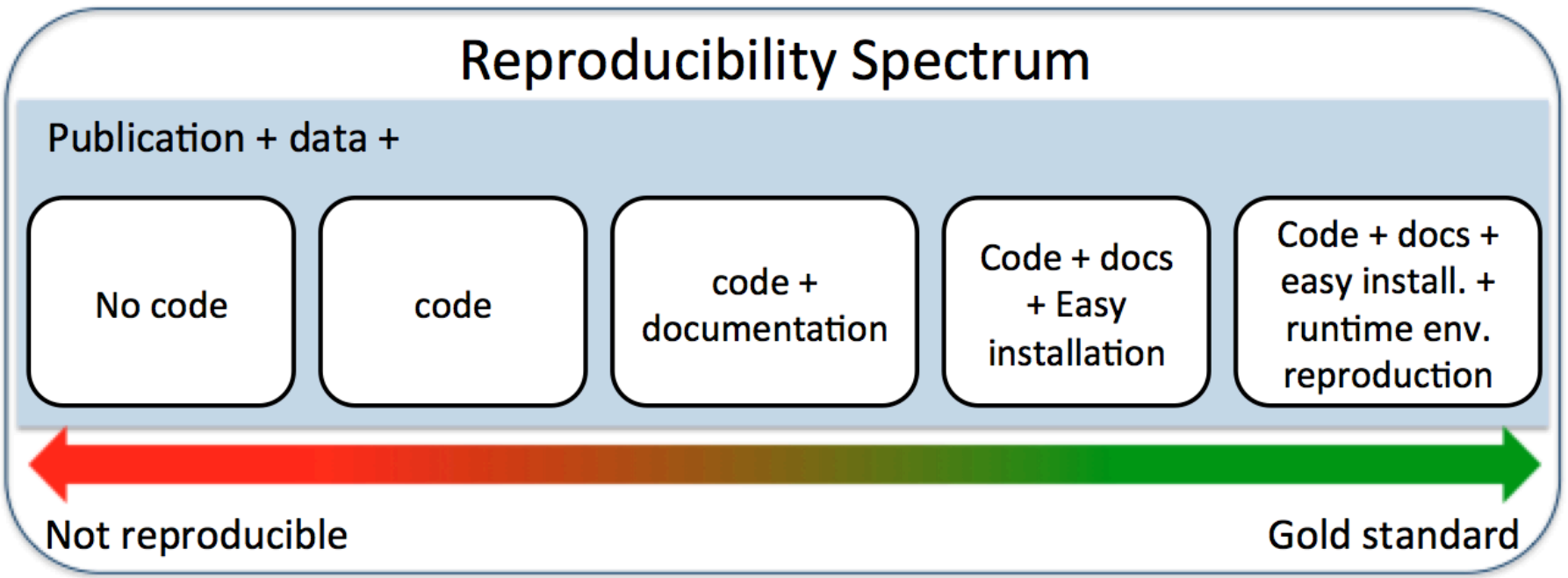
same data, same question (are dark-skinned players given more red cards?): odds ratios from 0.89 to 2.93, 20 teams significant, 9 not

Silberzahn et al. (2018)

⟶ shared data alone is not enough: code, environment and documentation are needed

Hardwicke et al. (2018) R Soc Open Sci; Obels et al. (2020) AMPPS; Silberzahn et al. (2018) AMPPS

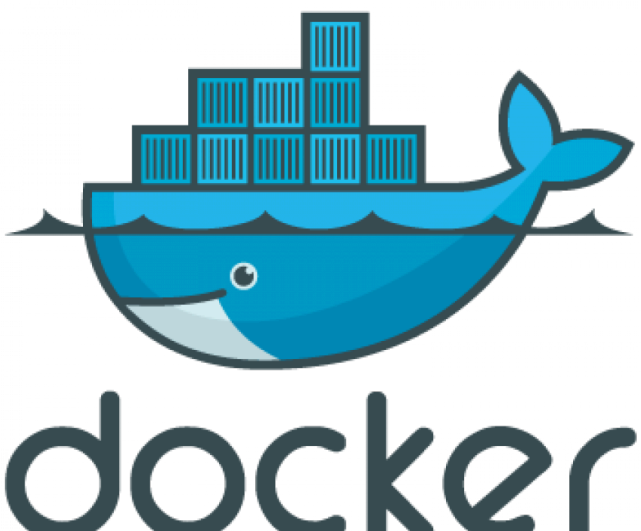
Computational Reproducibility



Adapted from Peng (2011) Science

Reproducibility and Dependencies

- high-level code has many dependencies
  - → Python version, package versions, external specialised software packages, system libraries, …
- sharing the environment is just as important as sharing the code
- Tools for the job:
  - → `conda` / `virtualenv` / `uv` (Python), `renv` (R)
  - → `pip freeze > requirements.txt`
  - → Docker
  - → …



Docker logo: Docker Inc.

Sharing platform: GitHub



- Platform for sharing code (hosts most open-source projects)
- Great for sharing scripts, materials and documentation
- Allows easy “forking” and adapting of existing projects
- Version control (`git`) built in
- Note: GitHub itself is *not* open source (owned by Microsoft); open alternatives: GitLab, Codeberg

Examples:

- github.com/ihrke/pypillometry
- github.com/ihrke/2016-placebo-tdcs-study
- github.com/ZsoltTuri/2021_rTMS_intensity_review_paper

Open Science Framework

- Easy sharing of materials, code and data
- Infrastructure for collaboration
- Facilities for pre-registration
- Integrates with other tools (e.g., GitHub)
- Integrates with preprint servers (e.g., PsyArXiv)

Examples:

- osf.io/43dp5 (example paper)
- osf.io/cv24f (Forskerlinje project, 2022)
- osf.io/wjvk2 (Master project, 2021)

OSF Projects is being retired (funding)

- **16 Nov 2026:** no new projects
- **19 Feb 2027:** existing projects become read-only

Still running: registrations (pre-registration), preprints, and existing public links and DOIs.

⟶ don't start a new project here

OSF logo: Center for Open Science (osf.io); retirement: COS announcement, Aug 2026

After OSF Projects: where to share code?



GitHub: develop and share code with version control (not an archive: repositories can be deleted)



Zenodo (CERN): free, permanent, a citable DOI for every version; link it to GitHub and each release is archived automatically



DataverseNO: Norwegian national research data repository, run by UiT; code + data together, curated



Software Heritage: archives all public source code; save your repository with one click

Logos: GitHub; Zenodo (CERN); DataverseNO (UiT); Software Heritage (public domain, Wikimedia Commons)

What does a shareable project look like?

```
my-thesis/
├── README.md
├── LICENSE
├── requirements.txt
├── data/
│   ├── raw/
│   └── processed/
├── code/
│   ├── 01_preprocess.py
│   ├── 02_analysis.py
│   └── run_all.py
└── output/
    ├── figures/
    └── tables/
```

- **README**: what is this, how do I run it?
- **LICENSE**: may I reuse it?
- **Environment** file: which versions?
- **Raw data stays raw**; every step is code
- **One script** reproduces all results
- Relative paths, no `C:\Users\...`
- Test: can a fellow student run it on *their* laptop?

AI coding assistants and open code



built on open source

- LLMs are trained on (mostly open-source) public code
- generated code may reproduce licensed code without attribution
- open questions: who is the author? which licence applies?

…and for reproducibility

- AI-written analysis code is still **your** code: you must understand and check it
- chat output is not reproducible: save the *code*, not the chat
- share the scripts as usual

AI itself

- **most AI is not open source**: models, training data and code stay closed (OpenAI, Google, Anthropic, …)
- **open-weight models** (Llama, Mistral, Qwen): run locally (training data not shared)

> **But:** AI makes code easier to share: READMEs, comments, tidy scripts, environment files.

Thanks for your attention!

Questions, comments, input?

contact me · matthias.mittner@uit.no

References

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